

2021 - Statistics - IV

Long Questions

- ① Employers want to know which days of the week employees are absent in a five-day work week. Most employers would like to believe that employees are absent equally during the week. Suppose a random sample of 60 managers were asked on which day of the week they had the highest number of employee absences. The results were distributed as in the table below. For the population of employees do the days for the highest member of absences occur with equal frequencies during a five-day work week? Test at a 5% significance level. Day of the week employees were most absent.

Number of absences	Monday	Tuesday	Wed	Thurs	Fri
	15	12	9	9	15

Solution

Hypothesis :-

H_0 : The number of absences occurs with equal frequencies during the five-day work week.

H_1 : The number of absences ^{does not} occurs with equal frequencies during the five-day work week.

Level of Significance:-

Test-statistics:-

$$\alpha = 0.05$$

PUAEP

$$\chi^2 = \sum \frac{(o_i - e_i)^2}{e_i}$$

Calculation:-

observed freq o_i	expected freq e_i	$o_i - e_i$	$(o_i - e_i)^2$	$\frac{(o_i - e_i)^2}{e_i}$
15	$\frac{60}{5} = 12$	3	9	0.75
12	12	0	0	0
9	12	3	9	0.75
9	12	3	9	0.75
15	12	3	9	0.75
Total 60	60	---	---	$\chi^2 = 3$

Critical region :-

$$\chi^2 \geq \chi^2_{0.05}(n-1)$$

$$\chi^2 \geq \chi^2_{0.05}(5-1)$$

$$\chi^2 \geq \chi^2_{0.05}(4)$$

$$9.49$$

Conclusion:-

Since the calculated value of Chi square (χ^2) is less than the critical region does not fall in critical region. So, the null hypothesis (H_0) is accepted and alternative hypothesis (H_1) is accepted.

- ② An education researcher is comparing four different algebra curricula. Eighth grade students are randomly assigned to one of the four groups. Their state achievement test scores are compared the end of the year. Use the appropriate statistical procedure to determine whether the curricula differ with respect to math achievement.

An alpha criterion of ~~0.05~~ 0.05 should be used for the test.

	n	Mean	sd
Curriculum 1	50	170.5	14.5
Curriculum 2	50	168.3	12.8
Curriculum 3	50	167.6	17.7
Curriculum 4	50	172.8	16.8

Solution:-

Hypothesis:-

H_0 : All the four Curricula are equal.

H_1 : Not all equal.

Level of significance:-

$$\alpha = 0.05$$

Test - statistics:-

$$F = \frac{S_b^2}{S_w^2}$$

Calculation:-

Calculate overall means

$$\text{Curriculum 1} = 170.5 \times 50 = 8525$$

$$\text{Curriculum 2} = 168.3 \times 50 = 8415$$

$$\text{Curriculum 3} = 167.6 \times 50 = 8380$$

$$\text{Curriculum 4} = 172.8 \times 50 = 8640$$

$$\text{overall mean} = \frac{8525 + 8415 + 8380 + 8640}{50 + 50 + 50 + 50}$$

$$= \frac{33960}{200}$$

$$\boxed{\text{overall mean } \bar{x} = 169.8}$$

$$\text{Between SS} = \sum n_i (u_i - \bar{u})^2$$

$$\text{Between SS} = 50(170.5 - 169.8)^2 + 50(168.3 - 169.8)^2 + 50(167.6 - 169.8)^2 + 50(172.8 - 169.8)^2$$

$$= 50(0.7)^2 + 50(1.5)^2 + 50(2.2)^2 + 50(3)^2$$

$$= 50(0.49) + 50(2.25) + 50(4.84) + 50(9)$$

$$= 24.5 + 112.5 + 242 + 450$$

$$\boxed{\text{Between SS} = 829}$$

$$\text{within SS} = \sum (n_i - 1) s_i^2$$

$$= (50-1)(14.5)^2 + (50-1)(12.8)^2 + (50-1)(17.7)^2 + (50-1)(16.8)^2$$

$$= 49(210.25) + 49(163.84) + 49(313.29) + 49(282.24)$$

$$= 10302.25 + 8028.16 + 15351.21 + 13829.76$$

$$\boxed{\text{within SS} = 47511.38}$$

Sources of Variation	d.f	SS	MS	Computed 'F'
	(k-1)			
Between	(4-1) 3	829	276.333	$F = 1.14$
Sample	(N-k)			
With in	200-4			
Sample	196	47511.38	242.405	- - -
Total	199	48340.38	- - -	- - -

Critical region:-

$$F \geq F_{0.05}(3, 196)$$

$$F \geq 2.60$$

$$1.14 \geq 2.60$$

Conclusion:-

Since the calculated value of F is (1.14) is less than the critical region or does not fall in critical region. So, the null hypothesis (H_0) is accepted.

③ Given the data

Fertilizer (x)	0.3	0.6	0.9	1.2	1.5	1.8	2.1	2.4
Corn Yield (y)	10	15	30	35	25	30	50	45

Assuming normality calculate the 95% Confidence interval for

- The value of α
- The value of β
- The true value of Y when $x=3.0$

Solution:-

$$\hat{y} = a + bx$$

$$b = \frac{n \sum xy - \sum x \sum y}{n \sum x^2 - (\sum x)^2}$$

	X	Y	X^2	Y^2	XY
	0.3	10	0.09	100	3
	0.6	15	0.36	225	9
	0.9	30	0.81	900	27
	1.2	35	1.44	1225	42
	1.5	25	2.25	625	37.5
	1.8	30	3.24	900	54
	2.1	50	4.41	2500	105
	2.4	45	5.76	2025	108
Total	10.8	240	18.36	8500	385.5

$$b = \frac{(8)(385.5) - (10.8)(240)}{(8)(18.36) - (10.8)^2}$$

$$b = \frac{3084 - 2592}{146.88 - 116.64}$$

$$b = \frac{492}{30.24}$$

$$b = 16.27$$

$$\hat{y} = a + bx$$

$$a = \bar{y} - b\bar{x}$$

$$a = \frac{240}{8} - (16.27) \left(\frac{10.8}{8} \right)$$

PUACP

$$= 30 - (16.27)(1.35)$$

$$= 30 - 21.9645$$

$$a = 8.0355$$

$$\hat{y} = a + bx$$

$$\hat{y} = 8.0355 + 16.27x$$

Find α

The 95% Confidence interval the α is given by

$$a \pm \frac{t_{\alpha/2}}{2} (n-2) S_a$$

$$S_a = S_{y.x} \sqrt{\frac{1}{n} + \frac{\sum \bar{x}^2}{\sum (x - \bar{x})^2}}$$

$$S_{y.x} = \sqrt{\frac{\sum (y - \bar{y})^2}{n - 2}}$$

$$= \sqrt{\frac{\sum y^2 - a \sum y - b \sum xy}{n - 2}}$$

$$= \sqrt{\frac{8500 - (9.04)(240) - (16.27)(385.5)}{6}}$$

$$= \sqrt{\frac{8500 - 1929.6 - 6272.685}{6}}$$

$$= \sqrt{\frac{298.315}{6}}$$

$$= \sqrt{49.719}$$

$$S_{y.x} = 7.05$$

$$\sum \bar{x}^2 = \left(\frac{10.8}{8}\right)^2 = (1.35)^2 = 1.8225$$

$$\sum \bar{x}^2 = 1.8225$$

$$\begin{aligned}
 \sum (x - \bar{x})^2 &= \sum x^2 - \frac{(\sum x)^2}{n} \\
 &= 18.36 - \frac{(10.8)^2}{8} \\
 &= 18.36 - \frac{116.64}{8} \\
 &= 18.36 - 14.58
 \end{aligned}$$

$$\boxed{\sum (x - \bar{x})^2 = 3.78}$$

So,

$$S_a = \frac{(7.05) \cdot 3.78}{8} = 3.225$$

$$= (7.05) \cdot 0.4625 = 3.260625$$

$$= (7.05) \cdot 0.6071$$

$$= (7.05) \cdot (0.779)$$

$$\boxed{S_a = 5.493}$$

$$a \pm \frac{t_{\alpha}(n-2) S_a}{2}$$

$$8.0355 \pm \frac{t_{0.05}(8-2)(5.493)}{2}$$

$$8.0355 \pm t_{0.025}(6)(5.493)$$

$$8.0355 \pm (2.447)(5.493)$$

$$8.0355 \pm 13.4413$$

$$8.0355 + 13.4413, 8.0355 - 13.4413$$

$$21.4748, -5.4078$$

Hence, the confidence interval for α is $(21.4748, -5.4078)$

ii) The 95% Confidence-interval the β is given

$$b \pm t_{\frac{\alpha}{2}(n-2)} S_b$$

$$S_b = \frac{S_{y|x}}{\sqrt{\sum (x - \bar{x})^2}}$$

$$= \frac{7.05}{\sqrt{3.78}}$$

$$= \frac{7.05 \text{PUACP} - 3.626}{1.9442}$$

$$S_b = 3.626$$

Substituting these values

$$16.27 \pm (2.447)(3.626)$$

$$16.27 \pm (8.88)$$

$$16.27 + 8.88, 16.27 - 8.88$$

$$25.15, 7.39$$

Hence the 95% Confidence interval The β is $(25.15, 7.39)$

iii) The 95% Confidence interval for the true of y when $x=3.0$ is Given:-

$$\hat{y} \pm t_{\frac{\alpha}{2}, (n-2)} S\hat{y}$$

$$\begin{aligned}\hat{y} &= a + bx_0 \quad \text{when } x=3.0 \\ &= 8.0355 + (16.27)(3.0) \\ &= 8.0355 + 48.81\end{aligned}$$

$$\boxed{\hat{y} = 56.85}$$

$$\begin{aligned}S\hat{y} &= S_{y.x} \sqrt{\frac{1}{n} + \frac{(x_0 - \bar{x})^2}{\sum (x - \bar{x})^2}} \\ &= (7.05) \sqrt{\frac{1}{8} + \frac{(3.0 - 1.35)^2}{3.78}}\end{aligned}$$

$$= (7.05) \sqrt{\frac{1}{8} + \frac{(1.65)^2}{3.78}}$$

$$= (7.05) \sqrt{\frac{1}{8} + \frac{2.7225}{3.78}}$$

$$= (7.05) \sqrt{0.125 + 0.72023}$$

$$= (7.05) \sqrt{0.84523}$$

$$= (7.05)(0.91936)$$

$$\boxed{S\hat{y} = 6.4814}$$

$$\hat{Y} \pm t_{\frac{\alpha}{2}(n-2)} S\hat{Y}$$

$$= (56.85) \pm (2.447)(6.4814)$$

$$= (56.85) \pm (15.86)$$

$$= 56.85 \pm 15.86$$

$$56.85 + 15.86, 56.85 - 15.86$$

$$72.71$$

$$40.99$$

Hence the 95% Confidence interval the true value of Y when $X = 3.0$ is $(72.71, 40.99)$.

2018

Long Questions

- ① Three Sections of the Same Statistics Course are taught by three teachers. The final grades given by the three teachers (independently) were recorded as follows:

Teacher A	75	91	84	45	82
Teacher B	59	83	99	77	65
Teacher C	66	77	51	90	73

Test the hypothesis that the average grades given by the three Teachers are same.

Solution:-

Hypothesis:-

$H_0: \mu_1 = \mu_2 = \mu_3$ all the three means are equal.

H_1 : All the three means are not equal.

Level of Significance:-

$$\alpha = 0.05$$

Test-Statistics:-

$$F = \frac{S_b}{\frac{P \times A \times P}{S_w}}$$

Calculation:-

	Teachers				
	A	B	C	T_i	X_{ij}^2
	75(5625)	59(3481)	66(4356)	200	13462
	91(8281)	83(6889)	77(5929)	251	21099
	84(7056)	99(9801)	51(2601)	234	19458
	45(2025)	77(5929)	90(8100)	212	16054
	82(6724)	65(4225)	73(5329)	220	16278
T_j	377	383	357	1117	86351
T_j^2	142129	146689	127449	416267	↓
X_{ij}^2	29711	30325	26315	86351	→ Checked

$$\begin{aligned}\text{Correction Factor (C.F)} &= \frac{\overline{1.}^2}{rK} \\ &= \frac{(1117)^2}{3 \times 5} \\ &= \frac{1247689}{15}\end{aligned}$$

$$\boxed{\text{Correction Factor (C.F)} = 83179.27}$$

$$\begin{aligned}\text{Total SS} &= \sum x_{ij}^2 - \text{C.F} \\ &= 86351 - 83179.27\end{aligned}$$

$$\boxed{\text{Total SS} = 3171.7333}$$

$$\begin{aligned}\text{Between SS} &= \sum \frac{T_j^2}{r} - \text{C.F} \\ &= \frac{416287}{5} - 83179.27 \\ &= 83253.4 - 83179.27\end{aligned}$$

$$\boxed{\text{Between SS} = 74.13}$$

$$\text{within SS} = \text{Total SS} - \text{Between SS}$$

$$= 3171.7333 - 74.13$$

$$\boxed{\text{within SS} = 3097.6033}$$

ANOVA Table

Sources of Variation	d.f	Sum of Square	Mean of Square	Computed F.
Between Teachers	$(K-1)$ $(3-1)$ 2	74.13	37.065	$F = 0.1436$
within Teachers	$N-K$ $15-3$ 12	3097.6033	258.133	
Total	14			

Critical region :-

$$F \geq F_{0.05}(2, 12)$$

$$F \geq 3.88$$

$$0.1436 < 3.88$$

Conclusion:-

Since the Calculated Value of F is less than the Critical region and does not fall in critical region. So, the null hypothesis (H_0) is accept and alternative hypothesis (H_1) is rejected.

- ② Is the type of beverage ordered with lunch at a restaurant independent of the age of the customer? A random poll of 55 lunch customers is taken, resulting in the following Contingency table of observed values. Use $\alpha = 0.01$ to determine whether the two variables are independent.

Age	preferred Beverage		
	Coffee/Tee	Soft Drink	Others(Milk)
1-34	8	14	9
35-55	5	4	2
>55	6	4	3

Solution:-

Hypothesis:-

H_0 : The type of beverage ordered is independent of the age of the customer.

H_1 : The type of beverage ordered is not independent of the age of the customer.

Level of Significance:-

$$\alpha = 0.01$$

Test-Statistics:-

$$\chi^2 = \frac{\sum (o_i - e_i)^2}{e_i}$$

Calculation:-

Age	Preferred Beverage			Total
	Coffe/Tea	Soft Drink	Other (Milk)	
1-34	8	14	9	31
35-55	5	4	2	11
755	6	14	3	13
Total	19	22	14	55

observed freq, o_i	expected freq, e_i	$(o - e_i)$	$(o - e_i)^2$	$\frac{(o - e_i)^2}{e_i}$
8	$\frac{19 \times 31}{55} = 10.71$	-2.71	7.3441	0.69
5	$\frac{19 \times 11}{55} = 3.8$	1.2	1.44	0.38
6	$\frac{19 \times 13}{55} = 4.49$	1.51	2.2801	0.51
14	$\frac{22 \times 31}{55} = 12.4$	1.6	2.56	0.21
4	$\frac{22 \times 11}{55} = 4.4$	-0.4	0.16	0.04
4	$\frac{22 \times 13}{55} = 5.2$	-1.2	1.44	0.28
9	$\frac{14 \times 31}{55} = 7.89$	1.11	1.2321	0.16
2	$\frac{14 \times 11}{55} = 2.8$	-0.8	0.64	0.23
3	$\frac{14 \times 13}{55} = 3.31$	-0.31	0.0961	0.03
$\sum o_i = 55$	$\sum e_i = 55$			$\chi^2 = 2.53$

Critical region:-

$$\chi^2 \geq \chi^2_{0.01} (r-1)(k-1)$$

$$\chi^2 \geq \chi^2_{0.01} (3-1)(3-1)$$

$$\chi^2 \geq \chi^2_{0.01} (2)(2)$$

$$\chi^2 \geq \chi^2_{0.01} (4)$$

$$\chi^2 \geq 13.28$$

$$2.53 \geq 13.28$$

Conclusion:-

Since the calculated value of χ^2 (2.53) is less than the critical region. So, the null hypothesis (H_0) is accepted and alternative hypothesis (H_1) is rejected.

PUACP

- ③ Five varieties of rice are compared in five randomized complete blocks. The SS for Total, Blocks and Varieties are 282.30, 26.96, and 155.44 respectively.

- i) Test the Significance of different between varieties means.
- ii) Compare the efficiency of this design with the design in which blocks are ignored.

Solution:-

Given that:-

$$\text{Total SS} = 282.30$$

$$\text{Block SS} = 26.96$$

$$\text{Varieties SS} = 155.44$$

$$\text{No. of varieties} = 5$$

$$\text{No. of blocks} = 5$$

Calculate the error SS

$$\begin{aligned}\text{Error SS} &= \text{Total SS} - (\text{BSS} + \text{VSS}) \\ &= 282.30 - (26.96 + 155.44) \\ &= 282.30 - 182.4 \\ &= 99.9\end{aligned}$$

$$\boxed{\text{Error SS} = 99.9}$$

Sources of variation	d.f	SS	MS	Computed F
Block SS	$\frac{(R-1)}{(5-1)} = 4$	26.96	6.74	---
Varieties SS	$\frac{4}{(5-1)(R-1)}$	155.44	38.86	$F = 6.224$
Error SS	16	99.9	6.244	---
Total	24	282.30	---	---

Hypothesis:-

H_0 : All the five varieties are equal.

H_1 : All the five varieties are not equal.

Level of Significance:-

$$\alpha = 0.05$$

Test - Statistics:-

$$F = \frac{MS_{\text{Varieties}}}{MS_{\text{Error}}}$$

Calculation:-

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Back page.

Critical region:-

$$F \geq F_{0.05}(4, 16)$$

$$F \geq 3.01$$

$$6.224 \geq 3.01$$

Conclusion:-

Since the calculated value is greater than the critical region. So, the null

hypothesis (H_0) is reject and alternative hypothesis (H_1) is accepted.

④ Given that

X_1	3	5	6	8	12	14
X_2	16	10	7	4	3	2
X_3	90	72	54	42	30	12

- i) Determine r_{12} , $r_{12.34}$ and $R_{1.23}$
ii) Test each one of these Correlation Coefficients for a Significance at 5% level.

Solution:- PUACP
calculate the means $\bar{X}_1$, $\bar{X}_2$ and $\bar{X}_3$.

$$\bar{X}_1 = \frac{3+5+6+8+12+14}{6} = \frac{48}{6} = 8$$

$$\bar{X}_2 = \frac{16+10+7+4+3+2}{6} = \frac{42}{6} = 7$$

$$\bar{X}_3 = \frac{90+72+54+42+30+12}{6} = \frac{300}{6} = 50$$

R_{1.23} Multiple Correlation Coefficient of X_1, X_2 , and X_3

$$R_{1.23} = \sqrt{\frac{Y_{12}^2 + Y_{13}^2 - 2Y_{12}Y_{13}Y_{23}}{1 - Y_{23}^2}}$$

Calculate Y_{12}, Y_{13}, Y_{23} and $Y_{12.34}$

$$Y_{12} = \frac{\sum (X_1 - \bar{X}_1)(X_2 - \bar{X}_2)}{\sqrt{\sum (X_1 - \bar{X}_1)^2 \cdot \sum (X_2 - \bar{X}_2)^2}}$$

$$\begin{aligned} \sum (X_1 - \bar{X}_1)(X_2 - \bar{X}_2) &= (-5)(9) + (-3)(3) + (-2)(0) \\ &\quad (0)(-3) + (4)(-4) + (6)(2) \\ &= -45 - 9 - 0 - 12 + 0 + 12 \end{aligned}$$

$$\boxed{\sum (X_1 - \bar{X}_1)(X_2 - \bar{X}_2) = -100}$$

PUACP

$$\sum (X_1 - \bar{X}_1)^2 = 25 + 9 + 4 + 0 + 16 + 36$$

$$\boxed{\sum (X_1 - \bar{X}_1)^2 = 90}$$

$$\sum (X_2 - \bar{X}_2)^2 = 81 + 9 + 9 + 0 + 16 + 25$$

$$\boxed{\sum (X_2 - \bar{X}_2)^2 = 140}$$

$$Y_{12} = \frac{-100}{\sqrt{90 \times 140}} = \frac{-100}{\sqrt{12600}}$$

$$Y_{12} = \frac{-100}{112.49}$$

$$\boxed{Y_{12} = -0.891}$$

$$r_{13} = \frac{\sum (X_1 - \bar{X}_1)(X_3 - \bar{X}_3)}{\sqrt{\sum (X_1 - \bar{X}_1)^2 \cdot \sum (X_3 - \bar{X}_3)^2}}$$

$$\begin{aligned}\sum (X_1 - \bar{X})(X_3 - \bar{X}_3) &= (-5)(40) + (-3)(22) + (-2)(4) \\ &\quad + (0)(-8) + (4)(-20) + (6)(-38) \\ &= -200 - 66 - 8 + 0 - 80 - 228\end{aligned}$$

$$\boxed{\sum (X_1 - \bar{X})(X_3 - \bar{X}_3) = -582}$$

$$\begin{aligned}\sum (X_3 - \bar{X}_3)^2 &= (40)^2 + (22)^2 + (4)^2 + (-8)^2 + \\ &\quad (-20)^2 + (-38)^2 \\ &= 1600 + 484 + 16 + 64 + 400 + 1444\end{aligned}$$

$$\boxed{\sum (X_3 - \bar{X}_3)^2 = 4008}$$

$$r_{13} = \frac{-582}{\sqrt{90 \times 4008}}$$

$$= \frac{-582}{\sqrt{360720}}$$

$$= \frac{-582}{600.6}$$

$$r_{13} = -0.969$$

$$r_{23} = \frac{\sum (x_2 - \bar{x}_2)(x_3 - \bar{x}_3)}{\sqrt{\sum (x_2 - \bar{x}_2)^2 \cdot \sum (x_3 - \bar{x}_3)^2}}$$

$$\begin{aligned} \sum (x_2 - \bar{x}_2)(x_3 - \bar{x}_3) &= (9)(40) + (3)(22) \\ &+ (0)(4) + (-3)(-8) + (-4)(-20) + (-5)(-38) \\ &= 360 + 66 + 24 + 80 + 190 \end{aligned}$$

$$\sum (x_2 - \bar{x}_2)(x_3 - \bar{x}_3) = 720$$

$$\sum (x_2 - \bar{x}_2)^2 = 140$$

$$\sum (x_3 - \bar{x}_3)^2 = 4008$$

$$* r_{23} = \frac{720}{\sqrt{140 \times 4008}}$$

$$r_{23} = \frac{720}{\sqrt{561120}}$$

$$r_{23} = \frac{720}{749.08}$$

$$\boxed{r_{23} = 0.9611}$$

$$r_{12.34} = \frac{r_{12} - r_{13}r_{23}}{\sqrt{(1-r_{13}^2)(1-r_{23}^2)}}$$

$$r_{12} = -0.891$$

$$r_{13} = -0.969$$

$$r_{23} = 0.9611$$

$$r_{12.34} = \frac{-0.891 + (0.969)(0.9611)}{\sqrt{(1-0.938961)(1-0.924)}}$$

$$= \frac{-0.891 + 0.9313}{\sqrt{(0.061)(0.076)}}$$

$$= \frac{0.0403}{\sqrt{0.004636}}$$

$$= \frac{0.0403}{0.0681}$$

$$\boxed{r_{12.34} = 0.5874}$$

$$R_{1.23} = \sqrt{\frac{r_{12}^2 + r_{13}^2 - 2r_{12}r_{13}r_{23}}{1 - r_{23}^2}}$$

$$r_{12} = -0.891$$

$$r_{13} = -0.969$$

$$r_{23} = 0.9611$$

$$r_{12}^2 = 0.794$$

$$r_{13}^2 = 0.938961$$

$$r_{23}^2 = 0.924$$

$$= \sqrt{\frac{0.794 + 0.938961 - 2(-0.891)(-0.969)(0.9611)}{1 - 0.924}}$$

$$= \sqrt{\frac{1.733 - 2(0.82979)}{0.076}}$$

$$= \sqrt{\frac{1.733 - 1.6596}{0.076}}$$

$$= \sqrt{\frac{0.07341}{0.076}}$$

$$= \sqrt{0.96595}$$

$$R_{1.23} = 0.9828$$

2017 Long Questions

- ① The weights of a random sample of 10 boxes of a particular brand of Cereal are 14.2, 13.7, 14.1, 14.3, 14.1, 13.8, 14.4, 14.8, 13.9 and 14.3. Test the hypothesis that $H_0: \sigma^2 = 0.02$ against the alternative $H_1: \sigma^2 < 0.02$ using a 0.01 level of significance.
Solution.

Hypothesis:-

$$H_0: \sigma^2 = 0.02$$

$$H_1: \sigma^2 < 0.02$$

Level of significance:-

$$\alpha = 0.01$$

Test-statistics:-

$$\chi^2 = \frac{ns^2}{\sigma^2}$$

Calculation:-

$$\chi^2 = \frac{ns^2}{\sigma^2} = \frac{n \sum (x - \bar{x})^2}{\sigma_0^2}$$

$$\sum (x - \bar{x})^2 = \sum x^2 - \frac{(\sum x)^2}{n}$$

$$\sum x^2 = (14.2)^2 + (13.7)^2 + (14.1)^2 + (14.3)^2 + (14.1)^2 + (13.8)^2 + (14.4)^2 + (14.8)^2 + (13.9)^2 + (14.3)^2$$

$$= 201.64 + 187.69 + 198.81 + 204.49 + 198.81 + 190.44 + 207.36 + 219.04 + 193.21 + 204.49$$

$$= 2005.98$$

$$\frac{(\sum x)^2}{n} = \frac{(14.2 + 13.7 + 14.1 + 14.3 + 14.1 + 13.8 + 14.4 + 14.8 + 13.9 + 14.3)^2}{10}$$

$$= \frac{(141.6)^2}{10} = \frac{20050.56}{10}$$

$$= ~~1416~~ 2005.056$$

$$\sum (x - \bar{x})^2 = \sum x^2 - \frac{(\sum x)^2}{n}$$

$$= 2005.98 - 2005.056$$

$$\boxed{\sum (x - \bar{x})^2 = 0.924}$$

$$\begin{aligned} \chi^2 &= \frac{\sum (x - \bar{x})^2}{\sigma^2} \\ &= \frac{0.924}{0.02} \end{aligned}$$

$$\chi^2 = 46.2$$

Critical region:-

$$\chi^2 \geq \chi^2_{1-\alpha}(n-1)$$

$$\chi^2 \geq \chi^2_{1-0.01}(10-1)$$

$$\chi^2 \geq \chi^2_{0.99}(9)$$

$$\chi^2 \geq 2.09$$

$$46.2 \geq 2.09$$

Conclusion:-

Since the calculated value of χ^2 is greater than the critical region or fall in critical region. So, the null hypothesis (H_0) is reject Accept (H_1).

② Three independent samples given the following results.

Size	observations
5	34, 40, 47, 60, 84
9	40, 54, 60, 67, 86, 92, 95, 98, 108.
3	48, 93, 100.

Use Bartlett's test to test the hypothesis of equal variances. Let $\alpha = 0.05$.

Solution:-

Hypothesis:-

$H_0: \sigma_1^2 = \sigma_2^2 = \sigma_3^2$ all the three variances are equal.

H_1 : All the three variance are not equal.

Level of significance:-

$$\alpha = 0.05$$

Test-statistics:-

$$U = \frac{2.306 (V \log_{10} S_p^2 - \sum V_i \log_{10} S_i^2)}{1 + \frac{1}{3(k-1)} \left[\sum_{i=1}^k \frac{1}{V_i} - \frac{1}{V} \right]}$$

Calculation:-

Bartlett's Test

$$H = \frac{2.3026 (V \log_{10} S_p^2 - \sum_{i=1}^k v_i \log_{10} S_i^2)}{1 + \frac{1}{3(k-1)} \left[\sum_{i=1}^k \frac{1}{v_i} - \frac{1}{V} \right]}$$

First we solve $V \log_{10} S_p^2$

$$S_p^2 = \frac{v_1 S_1^2 + v_2 S_2^2 + v_3 S_3^2}{v_1 + v_2 + v_3}$$

Compute S_1^2 , S_2^2 and S_3^2 .

Given that

Sample 1 = 34, 40, 47, 60, 84 ($n_1 = 5$)

Sample 2 = 40, 59, 60, 67, 86, 92, 95, 98, 108 ($n_2 = 9$)

Sample 3 = 46, 93, 100 ($n_3 = 3$)

Calculate the mean of each sample.

$$\bar{x}_1 = \frac{\sum(x_1)}{n} = \frac{34+40+47+60+84}{5}$$

$$\bar{x}_1 = \frac{265}{5}$$

$$\boxed{\bar{x}_1 = 53}$$

$$\bar{u}_2 = \frac{\sum u_2}{n}$$

$$= \frac{40+59+60+67+86+92+95+98+108}{9}$$

$$= \frac{705}{9}$$

$$\bar{u}_2 = 78.33$$

$$\bar{u}_3 = \frac{\sum u_3}{n}$$

$$= \frac{46+93+100}{3}$$

$$= \frac{239}{3}$$

$$\bar{u}_3 = 79.67$$

Calculate the variance of each sample.

$$S_i^2 = \frac{\sum (x_i - \bar{x}_i)^2}{n-1}$$

$$= \frac{(34-53)^2 + (40-53)^2 + (47-53)^2 + (60-53)^2 + (84-53)^2}{5-1}$$

$$= \frac{(-19)^2 + (-13)^2 + (-6)^2 + (7)^2 + (31)^2}{4}$$

$$= 361 + 169 + 36 + 49 + 961$$

$$= \frac{1576}{4}$$

$$S_1^2 = 394$$

$$S_2^2 = \frac{\sum (X_2 - \bar{X}_2)^2}{n-1}$$

$$= (40 - 78.33)^2 + (59 - 78.33)^2 + (60 - 78.33)^2 + (67 - 78.33)^2 + (86 - 78.33)^2 + (92 - 78.33)^2 + (95 - 78.33)^2 + (98 - 78.33)^2 +$$

$$108 - 78.33$$

$$9 - 1$$

$$= (-38.33)^2 + (-19.33)^2 + (-18.33)^2 + (-11.33)^2 + (7.67)^2 + (13.67)^2 + (16.67)^2 + (19.67)^2 + (29.67)^2$$

$$8$$

$$= 1469.189 + 128.3689 + 373.6489 + 335.9889 + 58.8289 + 18.8689 + 277.889 + 326.9089 + 386.9089 + 880.3089$$

$$8$$

$$= \frac{4098.002}{8}$$

$$S_2^2 = 512.250025$$

$$S_3^2 = \frac{\sum (X_3 - \bar{X}_3)^2}{n-1}$$

$$= \frac{(46 - 79.67)^2 + (93 - 79.67)^2 + (100 - 79.67)^2}{3-1}$$

$$= \frac{(-33.67)^2 + (13.33)^2 + (20.33)^2}{2}$$

$$= \frac{1133.67 + 177.69 + 413.31}{2}$$

$$= \frac{1724.67}{2}$$

$$[S_3^2 = 862.335]^{PUACP}$$

Calculate v and $vi si^2$ for each sample

$$v_1 = n-1 = 5-1 = 4$$

$$v_2 = n-1 = 9-1 = 8$$

$$v_3 = n-1 = 3-1 = 2$$

$$vi si^2 = 4 \times 394 = 1576$$

$$v_2 S_2^2 = 8 \times 512.2500025 = 4098.0002$$

$$V_3 S_3^2 = 2 \times 862.335 = 172.467$$

All the values put in formula Sp^2 .

$$Sp^2 = \frac{V_1 S_1^2 + V_2 S_2^2 + V_3 S_3^2}{V_1 + V_2 + V_3}$$

$$= \frac{1576 + 4098.0002 + 1724.67}{4 + 8 + 2}$$

$$= \frac{7398.6702}{14}$$

$$\boxed{Sp^2 = 528.48}$$

Find the value of v

$$V = V_1 + V_2 + V_3$$

$$V = 4 + 8 + 2$$

$$\boxed{V = 14}$$

$$= V \log_{10} 528.48$$

$$= 14 \log_{10} 528.48$$

$$= 14 \times 27.230$$

$$V \log_{10} S_i^2 = 38.122$$

Calculate the value of

$$\sum_{i=1}^K V_i \log_{10} S_i^2$$

$$S_i^2 = S_1^2 + S_2^2 + S_3^2$$

$$\begin{aligned} \sum_{i=1}^K V_i \log_{10} S_i^2 &= V_1 \log_{10} S_1^2 + V_2 \log_{10} S_2^2 + V_3 \log_{10} S_3^2 \\ &= 4 \log_{10} (394) + 8 \log_{10} (512.250025) + 2 \log_{10} (862.335) \end{aligned}$$

$$= 4 \log_{10} (394) + 8 \log_{10} (512.250025) + 2 \log_{10} (862.335)$$

$$= 4(2.5955) + 8(2.7695) + 2(2.936)$$

$$= 10.382 + 21.679 + 5.872$$

$$= 37.93$$

$$\sum_{i=1}^K V_i \log_{10} S_i^2 = 37.93$$

Calculate the value of $\sum_{i=1}^K \frac{1}{V_i} = ?$

$$\sum_{i=1}^K \frac{1}{V_i} = \frac{1}{V_1} + \frac{1}{V_2} + \frac{1}{V_3}$$

$$= \frac{1}{4} + \frac{1}{8} + \frac{1}{2}$$

$$= 0.25 + 0.125 + 0.5$$

$$\boxed{\sum_{i=1}^K \frac{1}{V_i} = 0.875}$$

Calculate the value of $\frac{1}{V}$

$$\frac{1}{V} = \frac{1}{V_1 + V_2 + V_3}$$

$$= \frac{1}{4 + 8 + 2} = \frac{1}{14}$$

$$\boxed{\frac{1}{V} = 0.0714}$$

Substituting these values into the formula for μ .

$$\mu = \frac{(2.3026)[(38.122) - (37.93)]}{1 + \frac{1}{3(3-1)} [0.875 - 0.0714]}$$

$$= \frac{(2.3026)(0.192)}{1 + \frac{1}{9-3} (0.8036)}$$

$$= \frac{0.4420992}{1 + \frac{1}{6}(0.8036)}$$

$$= \frac{0.4420992}{1 + (0.167)(0.8036)}$$

$$= \frac{0.4420992}{1 + 0.1342012}$$

$$= \frac{0.4420992}{1.1342012}$$

$$\boxed{u = 0.3898}$$

Critical region:-

$$X^2 \geq X^2_{0.05}(k-1) \quad k=3$$

$$X^2 \geq X^2_{0.05}(3-1)$$

$$X^2 \geq X^2_{0.05}(2)$$

$$X^2 \geq X^2_{0.05}(2)$$

$$0.3898 \geq 5.99$$

Conclusion:-

Since the calculated value of u is less than the critical region. So, the null hypothesis (H_0) is accepted.

- ③ An instructor of mathematics wished to determine the relationship of grades on a final examination to grades on two quizzes given during the semester. Calling x_1, x_2 and x_3 the grades of a student on the first quiz, second quiz and final examination respectively, he made the following computation for a total of 120 students.

$$\begin{array}{lll} \bar{x}_1 = 6.8 & S_1 = 1.0 & r_{12} = 0.60 \\ \bar{x}_2 = 7.0 & S_2 = 0.8 & r_{13} = 0.70 \\ \bar{x}_3 = 7.4 & S_3 = 9.0 & r_{23} = 0.65 \end{array}$$

- a) Find the least-square regression equation of x_3 on x_1 and x_2
b) Estimate the final grades of two students who scored respectively (1) 9 and 7 and (2) 4 and 8 on the two quizzes.
c) Compute $R^2_{3.12}$.

Solution:-

(a)

Since the standard deviations and linear correlation coefficients are given. therefore the

Estimated regression equation X_1, X_2 and X_3 is.

$$\frac{X_3 - \bar{X}_3}{S_3} = \left(\frac{X_1 - \bar{X}_1}{S_1} \right) \left(\frac{Y_{13} - r_{12}Y_{23}}{1 - r_{12}^2} \right) + \left(\frac{X_2 - \bar{X}_2}{S_2} \right) \left(\frac{Y_{23} - r_{12}Y_{13}}{1 - r_{12}^2} \right)$$

put the values in formula.

$$\frac{X_3 - 7.4}{.9} = \left(\frac{X_1 - 6.8}{1.0} \right) \left(\frac{0.70 - (0.60)(0.65)}{1 - (0.60)^2} \right) + \left(\frac{X_2 - 70}{0.8} \right) \left(\frac{(0.65) - (0.60)(0.70)}{1 - (0.60)^2} \right)$$

$$= \left(\frac{X_1 - 6.8}{1.0} \right) \left(\frac{0.70 - 0.39}{1 - 0.36} \right) + \left(\frac{X_2 - 70}{0.8} \right) \left(\frac{0.65 - 0.42}{1 - 0.36} \right)$$

$$= \left(\frac{X_1 - 6.8}{1.0} \right) \left(\frac{0.31}{0.64} \right) + \left(\frac{X_2 - 70}{0.8} \right) \left(\frac{0.23}{0.64} \right)$$

$$\frac{x_3 - 7.4}{9} = (x_1 - 6.8)(0.484) + (x_2 - 7.0)(0.449)$$

$$\boxed{x_3 - 7.4 = \frac{(x_1 - 6.8)(0.484)}{9} + \frac{(x_2 - 7.0)(0.449)}{9}}$$

$$= (x_1 - 6.8)$$

$$x_3 - 7.4 = (x_1 - 6.8)(0.484)(9) + (x_2 - 7.0)(0.449)(9)$$

$$x_3 - 7.4 = (x_1 - 6.8)(4.356) + (x_2 - 7)(4.041)$$

$$x_3 = 4.356x_1 - 29.6208 + x_2(4.041) - 28.287 + 7.4$$

$$x_3 = 4.356x_1 - 29.6208 + 4.041x_2 - 28.287 + 7.4$$

$$\boxed{x_3 = 4.356x_1 + 4.041x_2 + 16.09}$$

⑥

$$x_1 = 4, x_2 = 8$$

$$x_1 = 9, x_2 = 7$$

Student # 1

$$x_3 = 4.356(9) + 4.041(7) + 16.09$$

$$= 39.204 + 28.287 + 16.09$$

$$\boxed{x_3 = 83.581}$$

Student # 2

$$x_3 = 4.356(4) + 4.041(8) + 16.09$$

$$= 17.424 + 32.328 + 16.09$$

$$\boxed{x_3 = 65.842}$$

2) $R_{3.12} = ?$

$$R_{3.12} = \sqrt{\frac{\gamma_{13}^2 + \gamma_{23}^2 - 2\gamma_{12}\gamma_{13}\gamma_{23}}{1 - \gamma_{12}^2}}$$

$$= \sqrt{\frac{(0.70)^2 + (0.65)^2 - 2(0.60)(0.65)(0.70)}{1 - (0.60)^2}}$$

$$= \sqrt{\frac{0.49 + 0.4225 - 2(0.273)}{1 - 0.36}}$$

$$= \sqrt{\frac{0.49 + 0.4225 - 0.546}{0.64}}$$

$$= \sqrt{\frac{0.3665}{0.64}}$$

$$= \sqrt{0.5726}$$

$$\boxed{R_{3.12} = 0.757}$$

- ④ In a feeding experiment of some animals, the following results were obtained the numbers in the table being the gains in weight in pounds. The animals were in group of 3 each.

Groups	I	II	III	IV
A	7.0	16.0	10.5	13.5
B	14.0	15.5	15.0	21.0
C	8.5	16.5	9.5	13.5

Test the hypothesis of no difference in rations at $\alpha = 0.05$

Solution:-

Hypothesis:-

$H_0: \mu_1 = \mu_2 = \mu_3 = \mu_4$ all the four group means are equal

$H_1: \text{All the four group means are not equal}$

$H_0: \mu_A = \mu_B = \mu_C$ All the three ration mean are equal.

$H_1: \text{All the three ration}$

means are not equal.

Level of Significance:-

$$\alpha = 0.05$$

Test-Statistics:-

$$F_1 = \frac{S_1^2}{S_3^2}$$

$$F_2 = \frac{S_2^2}{S_3^2}$$

Calculation:-

Group 1							
Ration	I	II	III	IV	T _i	T _i ²	X _{ij} ²
A	7.0 (49)	16.0 (256)	10.5 (110.25)	13.5 (182.25)	47	2209	597.5
B	14.0 (196)	15.5 (240.25)	15.0 (225)	21.0 (441)	65.5	4290.25	1102.25
C	8.5 (72.25)	16.5 (272.25)	9.5 (90.25)	13.5 (182.25)	48	2304	617
T _j	29.5	48	35	48	160.5	8803.25	2316.75
T _j ²	870.25	2304	1225	2304	6703.25		↓
X _{ij} ²	317.25	768.5	475.5	805.5	2316.75	→	Checked

$$C.F = \frac{T^2}{rc} = \frac{(160.5)^2}{3 \times 4} = \frac{25760.25}{12}$$

$$[C.F = 2146.69]$$

$$\text{Total SS} = \sum X_{ij}^2 - C.F$$

$$= 2316.75 - 2146.69$$

$$[\text{Totalss} = 170.06]$$

$$\text{Between Group SS} = \frac{\sum T_j^2}{3} - C.F$$

$$= \frac{6703.25}{3} - 2146.69$$

$$= 2234.417 - 2146.69$$

$$[\text{Between Group SS} = 87.73]$$

PUACP

$$\text{Between Rations SS} = \frac{\sum T_i^2}{4} - C.F$$

$$= \frac{2203.25}{4} - 2146.69$$

$$= 2200.8125 - 2146.69$$

$$[\text{Between Rations} = 54.1225]$$

$$\text{Errorss} = \text{Totalss} - (\text{Groupss} + \text{Rationss})$$

$$= 170.06 - (87.73 + 54.1225)$$

$$= 170.06 - 141.8525$$

$$\boxed{\text{Error SS} = 28.2075}$$

ANOVA Table

Sources of variation	d.f	SS	MS	Computed F
Between Group	$k-1$ 3	87.73	29.24	$F_1 = 6.22$
Between Ration	$r-1$ 2	54.1225	27.06	$F_2 = 5.76$
Error	$(r-1)(k-1)$ 6	28.2075	4.70	---
Total	11	170.06	---	---

Critical region:-

$$F_1 \geq F_{0.05}(3,6)$$

4.76

$$F_2 \geq F_{0.05}(2,6)$$

5.14

Conclusions:-

Since the calculated value of F_1 and F_2 is greater than the critical region. So, the null hypothesis (H_0) is reject.

⑤ Carry out the analysis of variance for the following Latin Square.

$V_1 (2.3)$	$V_2 (3.0)$	$V_3 (3.3)$	$V_4 (2.5)$
$V_2 (3.1)$	$V_3 (4.1)$	$V_4 (2.4)$	$V_1 (2.4)$
$V_3 (4.3)$	$V_4 (2.5)$	$V_1 (2.1)$	$V_2 (2.9)$
$V_4 (2.6)$	$V_1 (2.0)$	$V_2 (2.4)$	$V_3 (4.4)$

Solution :-

Hypothesis :-

H_0 : There is no significant difference in the treatment means.

H_1 : There is a significant difference in the treatment means.

Level of significance:-

$$\alpha = 0.05$$

Test - Statistics:-

$$F = \frac{MST}{MSE}$$

Calculation:-

	columns			
Rows	1	2	3	4
1	2.3V ₁	3.0V ₂	3.3V ₃	2.5V ₄
2	3.1V ₂	4.1V ₃	2.4V ₄	2.4V ₁
3	4.3V ₃	2.5V ₄	2.1V ₁	2.9V ₂
4	2.6V ₄	2.0V ₁	2.4V ₂	4.4V ₃

	Columns				R _i	R _i ²	*
Rows	1	2	3	4	T _i	T _i ²	X _{ij} ²
1	2.3 (5.29)	3.0 (9)	3.3 (10.89)	2.5 (6.25)	11.1	123.21	31.43
2	3.1 (9.61)	4.1 (16.81)	2.4 (5.76)	2.4 (5.76)	11.0	121	37.94
3	4.3 (18.49)	2.5 (6.25)	2.1 (4.41)	2.9 (8.41)	11.8	139.24	37.56
4	2.6 (6.76)	2.0 (4)	2.4 (5.76)	4.4 (19.36)	11.4	129.96	35.88
C _i	T _j	12.3	11.6	10.2	12.2	46.3	536.41
C _i ²	T _j ²	151.29	134.56	104.04	148.84	538.73	↓
	X _{ij}	40.15	36.06	26.82	39.78	142.81	→ (checked)

$$C.F = \frac{T_{..}^2}{rc} = \frac{(46.3)^2}{4 \times 4} = \frac{2143.69}{16}$$

$$C.F = 133.981$$

$$\text{Total SS} = \sum x_{ij}^2 - C.F$$

$$= 142.8 - 133.981$$

$$\boxed{\text{Total SS} = 8.82}$$

$$\text{Row SS} = \frac{\sum T_i^2}{K} - C.F$$

$$= \frac{536.41}{4} - 133.981$$

$$= 134.1025 - 133.981$$

$$\boxed{\text{Row SS} = 0.1215}$$

$$\text{Columns SS} = \frac{\sum T_j^2}{h} - C.F$$

$$= \frac{538.73}{4} - 133.981$$

$$= 134.6825 - 133.981$$

$$\boxed{\text{Columns SS} = 0.7015}$$

For treatment totals, squares and means, we construct another table as follows:

Variety	$\bar{T}_h$	$\bar{T}_h^2$	Mean
V ₁	8.8	77.44	$\frac{8.8}{4} = 2.2$
V ₂	11.4	129.96	2.85
V ₃	16.1	259.21	4.025
Total V_h	10.0	100	2.5
Total.	46.3	566.61	---

$$\begin{aligned}
 \text{Treatment SS} &= \sum \frac{T_h^2}{K} - C.F \\
 &= \frac{566.61}{4} - 133.981 \\
 &= 141.6525 - 133.981
 \end{aligned}$$

$$\text{Treatment SS} = 7.6715$$

$$\begin{aligned}
 \text{Error SS} &= TSS - (\text{Rows} + \text{Columns SS} + \text{Treatment SS}) \\
 &= 8.82 - (0.12 + 0.714 + 7.6715) \\
 &= 8.82 - 8.5055
 \end{aligned}$$

$$\text{Error SS} = 0.3145$$

ANOVA Table.

Sources of Variation	d.f	SS	MS	Computed F
Rows	$4-1=3$	0.12	0.04	
Columns	$4-1=3$	0.714	0.23	
Treatments	3	7.6715	2.557	48.79
Error	$(K-1)(K-2)$ 6	0.3145	0.0524	
Total	15			

Critical region:-

$$F_{\text{cal}} \geq F_{0.05}(3, 6)$$

$$48.79 \geq 4.76$$

Conclusion :-

Since the calculated value of F is greater than the critical region. So the null hypothesis (H_0) is reject.